

# Simple Random Walk based on Generating Functions

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Let  $X_1, X_2, \dots$  be i.i.d. Bernoulli r.v. with values s.t.

$$\mathbb{P}(X_i = 1) = \mathbb{P}(X_i = -1) = \frac{1}{2}$$

We set  $S_0 = 0$  and

$$S_n = \sum_{i=1}^n X_i$$

Notations:

$$\begin{aligned} T_0 &= \inf\{n \geq 1 : S_n = 0\} \\ p_0(n) &= \mathbb{P}(S_n = 0) \\ f_0(n) &= \mathbb{P}(T_0 = n) = \mathbb{P}(S_1 \neq 0, \dots, S_{n-1} \neq 0, S_n = 0) \end{aligned}$$

Now consider generating functions

$$\begin{aligned} P_0(s) &= \sum_{n=0}^{\infty} p_0(n) s^n \\ F_0(s) &= \sum_{n=1}^{\infty} f_0(n) s^n \end{aligned}$$

We now prove the following theorem:

*Thm* / Let  $S_n$  be the random walk with parameters  $p$  and  $q = 1 - p$ . Then

- $P_0$  and  $F_0$  satisfy

$$P_0(s) = 1 + P_0(s)F_0(s)$$

- $P_0$  verifies

$$P_0(s) = \frac{1}{(1 - 4pqs^2)^{1/2}}$$

- $F_0$  is given by

$$F_0(s) = 1 - (1 - 4pqs^2)^{1/2}$$

proof

We set

$$A = \{S_n = 0\} \quad B_k = \{T_0 = k\}$$

We have

$$A = A \cap \left( \bigcup_{k=1}^n B_k \right) = \bigcup_{k=1}^n (A \cap B_k)$$

Thus we have

$$\mathbb{P}(A) = \sum_{k=1}^n \mathbb{P}(A|B_k)\mathbb{P}(B_k)$$

i.e.

$$p_0(n) = \sum_{k=1}^n p_0(n-k)f_0(k), \quad \forall n \geq 1$$

$$p_0(0) = 1$$

Thus

$$P_0(s) = 1 + P_0(s)F_0(s)$$

Now we compute  $p_0(n)$ , clearly we have

$$p_0(n) = \begin{cases} 0, & n \equiv 1(mod 2) \\ C_n^{n/2}(pq)^{n/2}, & n \equiv 0(mod 2) \end{cases}$$

By

$$(1+x^2)^{-1/2} = \sum_{m=0}^{\infty} \frac{(2m)!}{(m!)^2} \left(-\frac{x}{4}\right)^m$$

We have

$$P_0(s) = \frac{1}{(1-4pqs^2)^{1/2}}$$

Everything follows then.

*Thm II* We have

$$\mathbb{P}(T_0 < \infty) = 1 - |p - q|$$

proof

$$\mathbb{P}(T_0 < \infty) = F_0(1)$$